第 10 章　NK 模型：时变与承诺

在本章，我们遵循通货膨胀目标制（inflation targeting），并选定 $\hat{\pi}_t = 0$，此时不存在产出缺口，即 $\hat{y}_t = \hat{y}_t^n$。

10.1　供给侧的无效率

10.2　承诺下的最优货币政策

中央银行一定会执行诺言，那么它面临如下最优化问题

$$
\begin{aligned}
\min_{\hat{x}_t, \hat{\pi}_t{}_{t=0}^{\infty}} \quad & \frac{1}{2} \mathrm{E}_0 \sum_{t=0}^{\infty} \beta^t \left(\hat{\pi}_t^2 + \alpha_x \hat{x}_t^2 \right) \\
\text{s.t.} \quad & \hat{\pi}_t = \beta \mathrm{E}_t \hat{\pi}_{t+1} + \kappa \hat{x}_t + \hat{u}_t \\
& \hat{u}_t = \rho_u \hat{u}_{t+1} + \varepsilon_t^u
\end{aligned}
$$

构建拉格朗日函数

$$
\mathscr{L} = \mathrm{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{1}{2} \left(\hat{\pi}_t^2 + \alpha_x \hat{x}_t^2 \right) + \gamma_t \left(\hat{\pi}_t - \kappa \hat{x}_t - \beta \hat{\pi}_{t+1} - \hat{u}_t \right) \right]
$$

计算偏导数获得一阶条件

$$
\begin{aligned}
\frac{\partial \mathscr{L}}{\partial \hat{x}_t} &= \alpha_x \hat{x}_t - \kappa \gamma_t = 0 \\
\frac{\partial \mathscr{L}}{\partial \hat{\pi}_t} &= \hat{\pi}_t + \gamma_t - \beta \gamma_{t-1} = 0
\end{aligned}
$$

结合两式消去 γ_t 得 $\{x_t\}$ 的差分方程并作迭代

$$
\begin{aligned}
\hat{x}_t &= \hat{x}_{t-1} - \frac{\kappa}{\alpha_x}\hat{\pi}_t = \hat{x}_{t-2} - \frac{\kappa}{\alpha_x}\left(\hat{\pi}_t + \hat{\pi}_{t+1}\right) = \cdots = \hat{x}_{-1} - \frac{\kappa}{\alpha_x}\left(\hat{\pi}_t + \hat{\pi}_{t-1} + \cdots + \hat{\pi}_0\right) \\
&= 0 - \frac{\kappa}{\alpha_x}\left[\left(\log P_t - \log P_{t-1}\right) + \left(\log P_{t-1} - \log P_{t-2}\right) + \cdots + \left(\log P_0 - \log P_{-1}\right)\right] \\
&= -\frac{\kappa}{\alpha_x}\left(\log P_t - \log P_{-1}\right) =: -\frac{\kappa}{\alpha_x}\check{P}_t
\end{aligned}
$$